

Three improved lower bounds for hypergraph Ramsey numbers

Explicit group-invariant colourings improving three entries of §7.1(a) of the dynamic survey *Small Ramsey Numbers*. Each bound is carried by an object small enough to print, not by a search log.

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$$R(4,4,4;3) \geq 84 \quad R(4,6;3) \geq 64 \quad R(5,5;4) \geq 36$$

Under the survey's convention that R is the *least* n forcing a monochromatic clique, a good colouring on n points proves $R \geq n+1$.

CELL	SURVEY REV. #18	THIS WORK	WITNESS
$R(4,4,4;3)$	≥ 79	≥ 84	83 points, 3 colours
$R(4,6;3)$	≥ 63	≥ 64	63 points, 2 colours
$R(5,5;4)$	≥ 35	≥ 36	35 points, 2 colours

The objects, not the search

Each result is a group-invariant colouring given as a small table over group orbits rather than as solver output. For $R(4,4,4;3)$: let $p = 83$, let $a = 9$ generate the order-41 subgroup of Z_{83}^* , and let $G = \{x \rightarrow a^i x + b\}$, of order 3403. G acts on the 91,881 triples of Z_{83} with exactly **27 orbits, each free of size 3403**; 27 numbers give the colours, nine per colour, so the three colour classes are equal at 30,627 each.

That equality is *not* forced. Each triple orbit has size 3403, so every invariant color class has size divisible by 3403; this does not force the color classes to have equal sizes. Enumerating every invariant good colouring of this class gives **36** of them — twelve splitting 9/9/9 and twenty-four splitting 8/9/10. The other two witnesses use 120 and 107

orbits, under actions that are not free. Every construction regenerates its whole colouring from its table alone, with no solver.

Verification

All twelve deposited witnesses are checked against their stated candidate universes, each under its own hash-bound run. For the three headline objects those universes contain **1,837,620** four-subsets at $n = 83$, **324,632** five-subsets at $n = 35$, and **595,665** four-subsets plus **67,945,521** six-subsets at $n = 63$. The $n=63$ verifier uses prefix pruning, so these are candidate-universe sizes rather than loop-body visit counts. For this $n = 63$ witness, colour 0 contains no monochromatic $K_4^{(3)}$, and colour 1 contains no monochromatic $K_6^{(3)}$. A separate reversed- K_4 control counts 102,627 colour-1 $K_4^{(3)}$ subgraphs. This is a discriminating control against certain implementation failures; it does not establish the colour convention or validate the K_6 branch. A separate six-vertex control assigns colour 1 to all twenty triples and requires the K_6 detector to report one violation.

Negative results carry certificates too: a separate RUP/RAT proof checker re-certifies five UNSAT symmetry classes of the $n = 63$ tower, and its discriminating control weakens a real instance until it becomes satisfiable and requires the checker to refuse the original proof.

An obstruction theorem

If an affine group satisfies $|G| = C(p, 2)$ and its multiplier subgroup has index two and omits -1 (equivalently, within that index-two family, $p \equiv 3 \pmod{4}$), it acts regularly on unordered pairs. Adjoining a fixed point then forces every triple through that point into a single orbit, and hence one colour; no base triple may take that colour, so for the diagonal problem the base must be coloured with $m-1$ colours while avoiding a monochromatic $K_4^{(3)}$ — impossible only once the base reaches or exceeds the $(m-1)$ -colour Ramsey number. **The threshold is a hypothesis, not a detail:** at $m = 3$ it is $R(4, 4; 3) = 13$ and both towers clear it; at $m = 4$ it is $R(4, 4, 4; 3)$ itself, and $p = 79$ does not. The order condition alone is insufficient when $p \equiv 1 \pmod{4}$: the corresponding fixed-point classes are satisfiable at $p = 13$ and $p = 37$. These examples target the omitted -1 condition and do not by themselves establish independence of every condition in a broader generalisation.

What is not claimed

- **Novelty is provisional.** Both incumbent constructions are *personal communications* (2018 and 2021) that the bibliographic sources searched could not supply. Nothing stronger was found, but a newer unpublished communication is not excluded.
- **No upper bound is established here.** The survey does not display a two-sided interval for these cells, so these improvements do not close a bracket.
- **The incumbent constructions were not re-verified here.** For the two [Dyb3] cells, the public-addendum files were inspected for their sizes, hashes and declared vertex counts, but were not re-verified as colourings. The [Ex24] construction underlying the DS1 entry for $R(5, 5; 4)$ was not obtained through the documented retrieval channels.
- **Every negative result is a statement about a symmetry class**, never about a Ramsey number — a class can be empty while the cell is wide open.

Reproducing

```
git clone https://github.com/SeverinVisionary/hypergraph-ramsey-bounds
cd hypergraph-ramsey-bounds
pip install -r requirements.txt
bash reproduce.sh --full
```

The full harness checks the listed construction, witness, obstruction and restricted-enumeration claims. It regenerates five selected UNSAT certifications; raw proof clauses are not included in this release. Literature comparisons and historical search summaries have separate evidence sources.

References

1. S. P. Radziszowski, *Small Ramsey Numbers*, Electronic Journal of Combinatorics, Dynamic Survey DS1, revision #18, 24 April 2026. [doi:10.37236/21](https://doi.org/10.37236/21)
2. H. Yang, *Three improved lower bounds for hypergraph Ramsey numbers: $R(4,4,4;3) \geq 84$, $R(4,6;3) \geq 64$, $R(5,5;4) \geq 36$ (v1.0)*, software and data, Zenodo, 2026. [doi:10.5281/zenodo.22698342](https://doi.org/10.5281/zenodo.22698342)

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